

ECO102: Introduction to Macroeconomics

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SECTION: 05

WEEK 01, LECTURE 02

CHAPTER 21: MEASURING GDP AND ECONOMIC GROWTH

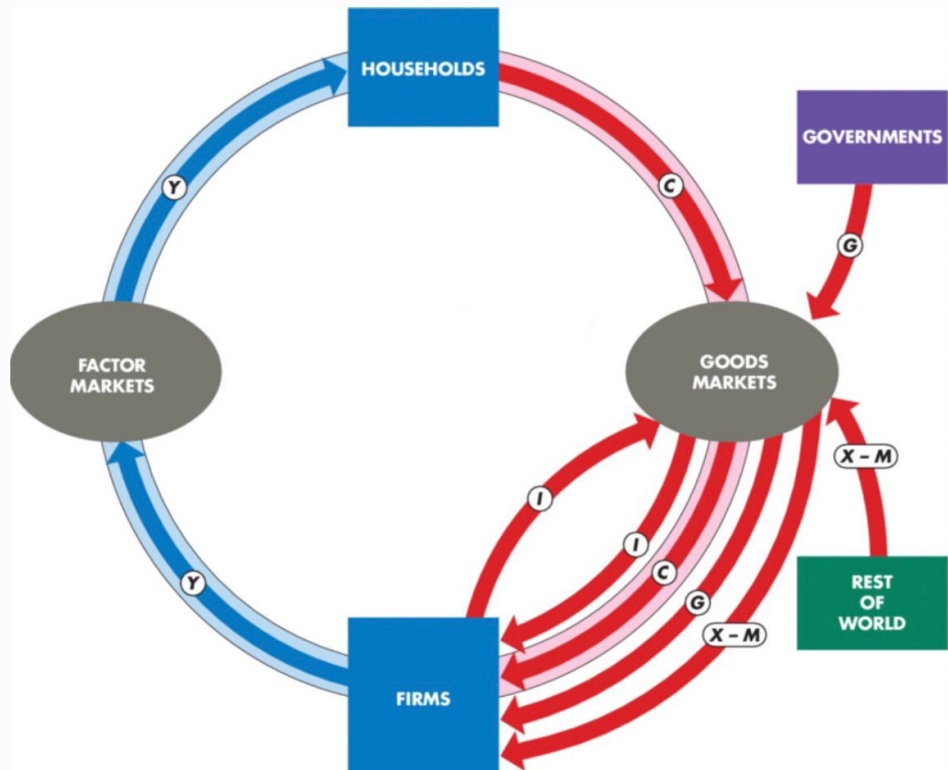
Gross Domestic Product

- ▶ **GDP** or **Gross Domestic Product** is the market value of all the final goods and services produced within a country in a given time period – usually a year.

This definition has four parts

- Market value
- Final goods and services
- Within a country
- In a given time period

Circular Flow of Expenditure and Income



- ▶ Household sells and firms buy the services of Labour, land and capital in **factor market**.
- ▶ Firms pay incomes (Y) to households.
- ▶ Households buy and firms sell consumption goods and services in the **goods market**.
- ▶ Households make *consumption expenditures* (C)
- ▶ Firms make *investments* (I)
- ▶ The government purchases goods and services (G)
- ▶ The rest of the world buys *net exports* ($X-M$)
- ▶ $X = \text{exports}$, $I = \text{Imports}$

GDP Equals Expenditure Equals Income

- ▶ GDP can be measured in 2 ways: Expenditure Approach, and Income Approach.
- ▶ **Expenditure Approach:** The total expenditure on goods and services – aggregate expenditure- equals consumption expenditure plus investment plus government expenditure plus net exports.

$$C + I + G + (X-M)$$

- ▶ **Income Approach:** Total income earned by producing goods and services- aggregate income- is equal to the total amount paid for the factors used- wages, interest, rent and profit.
- ▶ Because firms payout as incomes (including retained profits) everything they receive from the sale of their output, aggregate income equals aggregate expenditure.

$$GDP = Y = C + I + G + (X-M)$$

- ▶ GDP equals aggregate expenditure and equals aggregate income.

Measuring GDP : The Expenditure Approach

- ▶ The *expenditure approach* measures GDP as the sum of consumption expenditure (C), investment (I), government expenditure on goods and services (G), and net exports of goods and services ($X - M$).
- ▶ *Personal consumption expenditures* are the expenditures by households on goods and services produced in the country and in the rest of the world.
- ▶ *Gross private domestic investment* is expenditure on capital equipment and buildings by firms and the additions to business inventories. It also includes expenditure on new homes by households.
- ▶ *Government expenditure on goods and services* is the expenditure by all levels of government on goods and services, such as national defense and garbage collection.
- ▶ *Net exports of goods and services* are the value of exports minus the value of imports.

TABLE 21.1 GDP: The Expenditure Approach

Item	Symbol	Amount in 2010 (billions of dollars)	Percentage of GDP
Personal consumption expenditures	C	10,285	70.5
Gross private domestic investment	I	1,842	12.6
Government expenditure on goods and services	G	2,991	20.5
Net exports of goods and services	$X - M$	-539	-3.7
Gross domestic product	Y	14,579	100.0

Measuring GDP: The Income Approach

- ▶ The *income approach* measures GDP by summing the incomes that firms pay households for the services of the factors of production they hire—wages for labor, interest for capital, rent for land, and profit for entrepreneurship.
- ▶ The *National Income and Product Accounts* divide incomes into two big categories:
 1. Compensation of employees
 2. Net operating surplus
- ▶ Or in some editions it is: The Blue book incomes are grouped into three components:
 1. Compensation of employees
 2. Gross Operating Surplus
 3. Mixed Income

The Income Approach

- ▶ The sum of all the incomes is called net domestic income at factor cost.
- ▶ The term factor cost is used because it is the cost of factors of production used to produce final goods.
- ▶ When we sum the expenditures on final goods, we arrive at a total called *domestic product at market prices*.
- ▶ Market prices and factor cost diverge because of indirect taxes and subsidies.
- ▶ An **indirect tax** is a tax paid by consumers when they buy goods and services. A **subsidy** is a payment by the government to a producer.
- ▶ To get from factor cost to market price, we add indirect taxes and subtract subsidies.

TABLE 21.2 GDP: The Income Approach

Item	Amount in 2010 (billions of dollars)	Percentage of GDP
Compensation of employees	7,929	54.4
Net interest	924	6.3
Rental income	299	2.1
Corporate profits	1,210	8.3
Proprietors' income	1,050	7.2
<i>Net domestic income at factor cost</i>	11,412	78.3
Indirect taxes <i>less</i> subsidies	1,127	7.7
<i>Net domestic income at market prices</i>	12,539	86.0
Depreciation	1,860	12.8
GDP (income approach)	14,399	98.8
Statistical discrepancy	180	1.2
GDP (expenditure approach)	14,579	100.0

Nominal GDP and Real GDP

- ▶ **Real GDP** is the value of final goods and services produced in a given year when *valued at the prices of a reference base year*.
- ▶ **Nominal GDP** is the value of final goods and services produced in a given year when valued at the prices of that year.
- ▶ Nominal GDP is just a more precise name for GDP.
- ▶ GDP of Bangladesh in 2022 is USD 460.2 billion and 2010 it was USD 115.28 billion.
- ▶ To isolate the increase in production from the rise in prices, we distinguish between *real* GDP and *nominal* GDP.

Note: To see the GDP of Bangladesh in each category to this link:
https://mof.portal.gov.bd/sites/default/files/files/mof.portal.gov.bd/page/f2d8fabb_29c1_423a_9d37_cdb500260002/Chapter-2%20%28English-2023%29.pdf

Calculating Real GDP

TABLE 21.3 Calculating Nominal GDP and Real GDP

Item		Quantity (millions)	Price (dollars)	Expenditure (millions of dollars)
(a) In 2005				
C	T-shirts	10	5	50
I	Computer chips	3	10	30
G	Security services	1	20	20
Y	Real and Nominal GDP in 2005			100
(b) In 2010				
C	T-shirts	4	5	20
I	Computer chips	2	20	40
G	Security services	6	40	240
Y	Nominal GDP in 2010			300
(c) Quantities of 2010 valued at prices of 2005				
C	T-shirts	4	5	20
I	Computer chips	2	10	20
G	Security services	6	20	120
Y	Real GDP in 2010			160

- ▶ Table 21.3 shows the quantities produced and the prices in 2005 (the base year) and in 2010.
- ▶ In part (a), we calculate nominal GDP in 2005. For each item, we multiply the quantity produced in 2005 by its price in 2005 to find the total expenditure on the item.
- ▶ Because 2005 is the base year, both real GDP and nominal GDP equal \$100 million.
- ▶ In Table 21.3(b), we calculate nominal GDP in 2010, which is \$300 million. Nominal GDP in 2010 is three times its value in 2005. But by how much has production increased? Real GDP will tell us.
- ▶ For each item, we multiply the quantity produced in 2010 by its price in 2005. We then sum these expenditures to find real GDP in 2010, which is \$160 million.

The Uses of Real GDP

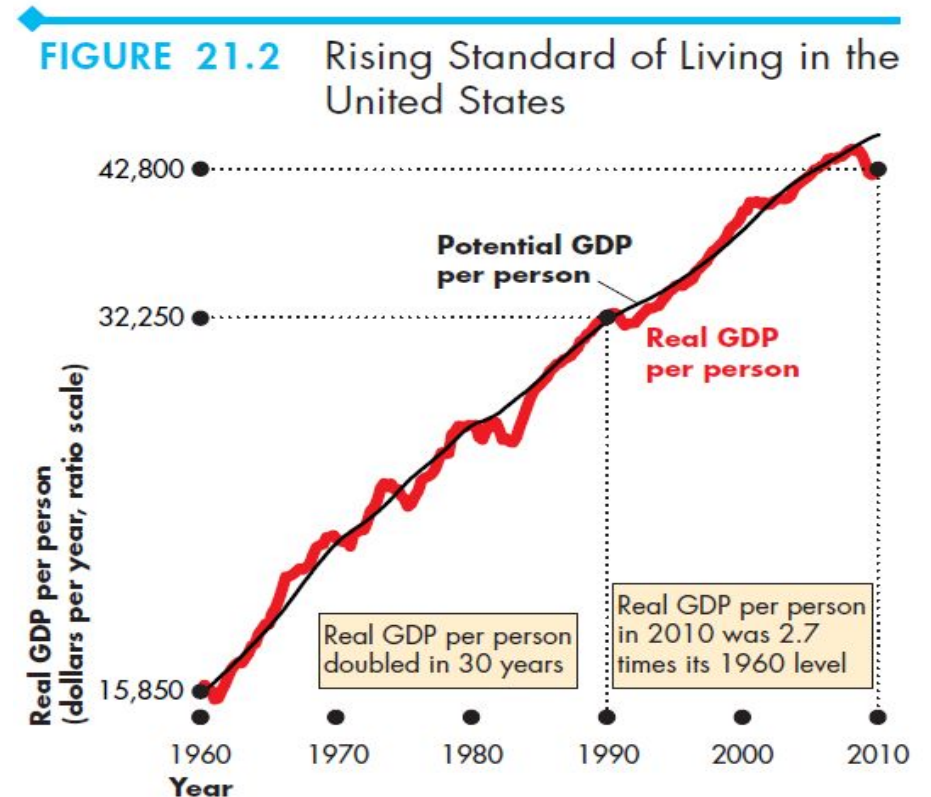
- ▶ Economists use estimates of real GDP for two main purposes:
 - To compare the standard of living over time
 - To compare the standard of living across countries

Standard of Living over time: One method of comparing is to calculate real GDP per person in different years.

- ▶ **Real GDP per person** is real GDP divided by the population. It tells us the value of goods and services that the average person can enjoy.
- ▶ **Long-Term Trend** A handy way of comparing real GDP per person over time is to express it as a ratio of some reference year.
- ▶ This trend tells us two features of our expanding living standard:
 - ▶ The growth of potential GDP
 - ▶ Fluctuations of real GDP per person

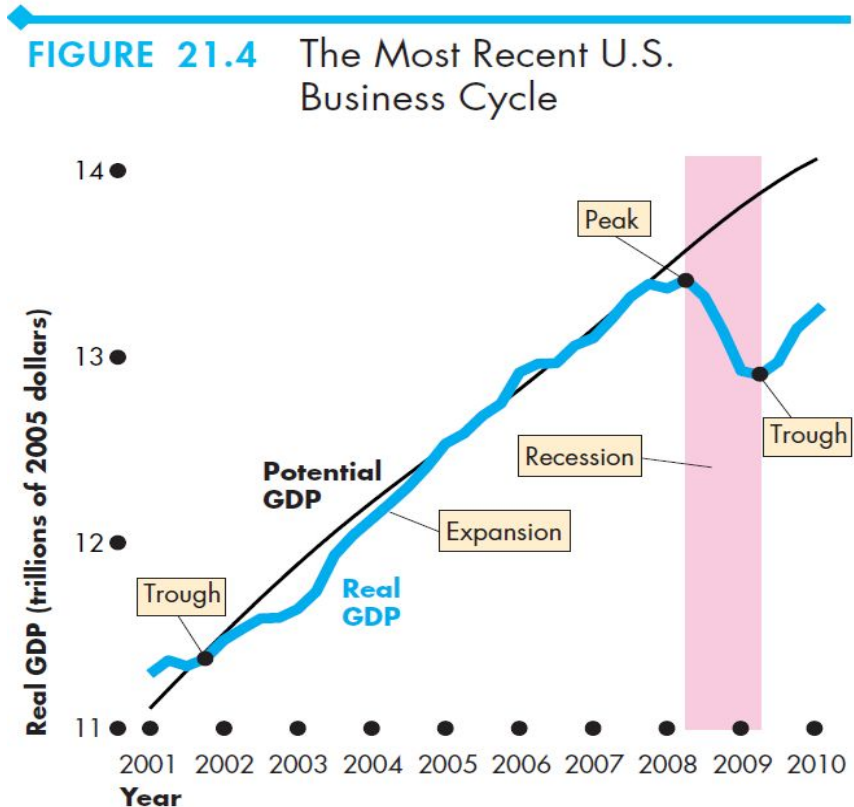
The Growth of Potential GDP

- **Potential GDP** is the value of production when all the economy's labor, capital, land and entrepreneurial ability are fully employed.



Real GDP Fluctuations- The Business Cycle

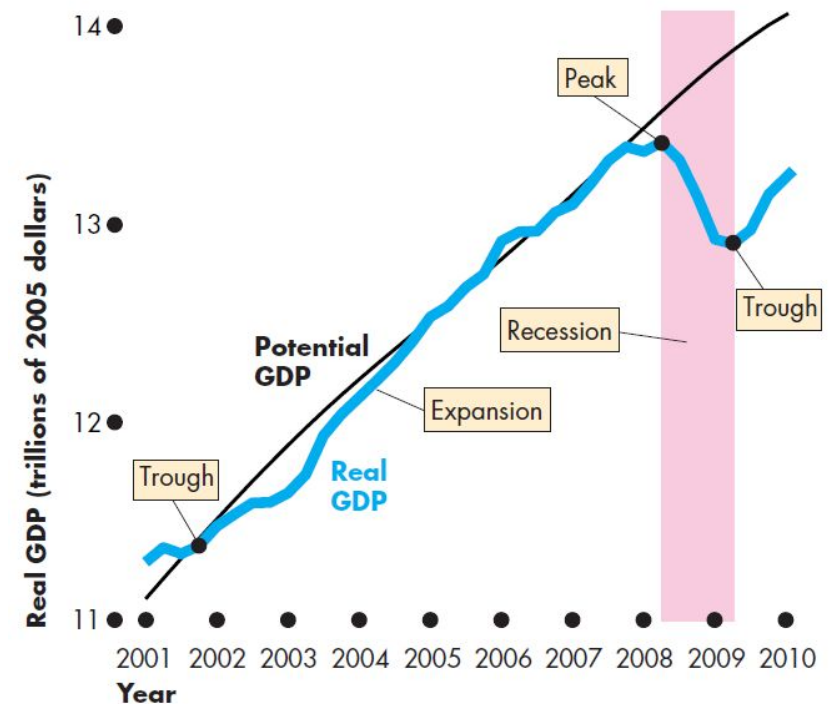
- ▶ The **business cycle** is a periodic but irregular up-and-down movement of total production and other measures of economic activity.
- ▶ The business cycle isn't a regular predictable cycle. Every cycle has two phases:
 1. Expansion
 2. Recession
- ▶ Two turning points:
 1. Peak
 2. Trough



The Business Cycle

- ▶ An **Expansion** is a period during which real GDP increases. In the early stage of an expansion real GDP returns to potential GDP and as the expansion progresses, potential GDP grows and real GDP eventually exceeds potential GDP.
- ▶ **Recession** is a period during which real GDP decreases—its growth rate is negative—for at least two successive quarters.
 - ▶ It is a period of significant decline in total output, income, employment, and trade, usually lasting from six months to a year, and marked by contractions in many sectors of the economy.
- ▶ An expansion ends and recession begins at a business cycle *peak*, which is the highest level that real GDP has attained up to that time. A recession ends at a *trough*, when real GDP reaches a temporary low point and from which the next expansion begins.

FIGURE 21.4 The Most Recent U.S. Business Cycle



Limitations of Real GDP (Self-study)

- ▶ Real GDP measures the value of goods and services that are bought in markets. Some of the factors that influence the standard of living and that are not part of GDP are
 - Household production
 - Underground economic activity
 - Health and life expectancy
 - Leisure time
 - Environmental quality
 - Political freedom and social justice